

NETWORK SECURITY ENGINEERING PORTFOLIO

Laboratory Report

Basic BGP Lab

Multi-Autonomous System Routing with Border Gateway Protocol

Prepared by

Timothy Otim Lutara

BSc Information Technology (Cybersecurity)
ISBAT University, Kampala, Uganda

Platform: EVE-NG Community Edition
May 2026

1. Project Overview

This report documents the design, configuration, and verification of a multi-Autonomous System Border Gateway Protocol (BGP) network built in EVE-NG. The lab consisted of 12 routers distributed across four distinct Autonomous Systems, each representing a separate network organization. The objective was to establish full end-to-end IP connectivity from AS64900 to AS64915, passing through two transit Autonomous Systems, using BGP as the inter-domain routing protocol.

BGP is the routing protocol that powers the internet. Unlike Interior Gateway Protocols such as OSPF and EIGRP which are designed for routing within a single organization, BGP is designed for routing between organizations. Each organization operates its own Autonomous System identified by a unique AS number. BGP enables these organizations to exchange routing information and establish reachability across the global internet.

This lab simulates that exact scenario in a controlled environment, making it directly relevant to real-world network engineering, internet service provider operations, and enterprise WAN design.

1.1 Lab Objectives

1. Configure all Serial and Ethernet interfaces with the correct IP addresses as specified in the lab addressing scheme.
2. Configure loopback interfaces on each router using the Y.Y.Y.Y/24 scheme to serve as stable BGP router identities.
3. Deploy OSPF as the Interior Gateway Protocol within each Autonomous System to enable loopback reachability for iBGP sessions.
4. Establish iBGP full-mesh peering within each Autonomous System using loopback addresses as the session source.
5. Configure eBGP peering across all Autonomous System boundaries using directly connected interface addresses.
6. Apply next-hop-self on all border routers to ensure correct route propagation to internal BGP peers.
7. Verify end-to-end connectivity from R1 in AS64900 to R12 in AS64915 using a loopback-sourced ping.

2. Network Topology

The lab topology consisted of four Autonomous Systems interconnected in a linear chain. AS64900 connected to AS64905, which connected to AS64910, which connected to AS64915. Traffic originating in AS64900 had to traverse two transit Autonomous Systems before reaching its destination in AS64915.

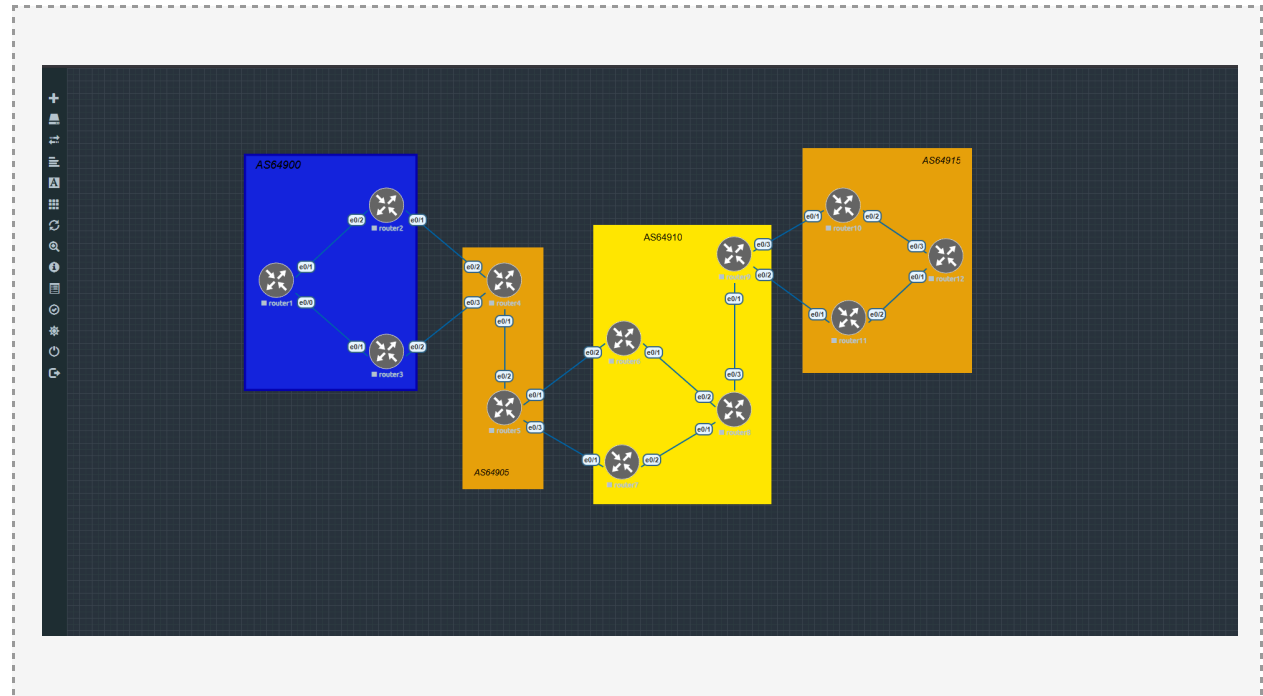


Figure 1: Full Lab Topology in EVE-NG

2.1 Autonomous System Breakdown

Autonomous System	AS Number	Routers	Role
AS64900	64900	R1, R2, R3	Origin AS
AS64905	64905	R4, R5	Transit AS
AS64910	64910	R6, R7, R8, R9	Transit AS
AS64915	64915	R10, R11, R12	Destination AS

2.2 eBGP Inter-AS Links

The following links cross Autonomous System boundaries and carry eBGP sessions between border routers.

Router A	Router B	Subnet	AS A	AS B
R2	R4	192.168.24.0/24	AS64900	AS64905
R3	R4	192.168.34.0/24	AS64900	AS64905
R5	R6	192.168.56.0/24	AS64905	AS64910
R5	R7	192.168.57.0/24	AS64905	AS64910
R9	R10	192.168.109.0/24	AS64910	AS64915
R9	R11	192.168.119.0/24	AS64910	AS64915

3. IP Addressing Design

The lab used a structured addressing scheme where each point-to-point link between two routers received its own dedicated subnet. The subnet number directly reflected the two router numbers involved in the link, making the addressing plan intuitive and easy to troubleshoot.

3.1 Point-to-Point Link Addressing Scheme

All point-to-point links used the 192.168.YY.x/24 scheme where YY represents the concatenation of the two connected router numbers. The lower-numbered router always received the .1 host address and the higher-numbered router always received the .2 host address.

Link	Subnet	Lower Router (.1)	Higher Router (.2)	AS
R1 to R2	192.168.12.0/24	R1: 192.168.12.1	R2: 192.168.12.2	AS64900
R1 to R3	192.168.13.0/24	R1: 192.168.13.1	R3: 192.168.13.2	AS64900
R2 to R4	192.168.24.0/24	R2: 192.168.24.1	R4: 192.168.24.2	eBGP
R3 to R4	192.168.34.0/24	R3: 192.168.34.1	R4: 192.168.34.2	eBGP
R4 to R5	192.168.45.0/24	R4: 192.168.45.1	R5: 192.168.45.2	AS64905
R5 to R6	192.168.56.0/24	R5: 192.168.56.1	R6: 192.168.56.2	eBGP
R5 to R7	192.168.57.0/24	R5: 192.168.57.1	R7: 192.168.57.2	eBGP
R6 to R8	192.168.68.0/24	R6: 192.168.68.1	R8: 192.168.68.2	AS64910
R7 to R8	192.168.78.0/24	R7: 192.168.78.1	R8: 192.168.78.2	AS64910
R8 to R9	192.168.89.0/24	R8: 192.168.89.1	R9: 192.168.89.2	AS64910
R9 to R10	192.168.109.0/24	R9: 192.168.109.1	R10: 192.168.109. 2	eBGP
R9 to R11	192.168.119.0/24	R9: 192.168.119.1	R11: 192.168.119.2	eBGP

R10 to R12	192.168.120.0/24	R10: 192.168.120.1	R12: 192.168.120. 2	AS64915
R11 to R12	192.168.121.0/24	R11: 192.168.121.1	R12: 192.168.121. 2	AS64915

3.2 Loopback Interface Addressing

Each router was assigned a loopback interface using the Y.Y.Y.Y/24 scheme, where Y is the router number. Loopback interfaces are virtual interfaces that exist entirely in software and never go down as long as the router is operational. This makes them ideal as stable identities for BGP sessions.

Router	Loopback Address	Router	Loopback Address
R1	1.1.1.1/24	R7	7.7.7.7/24
R2	2.2.2.2/24	R8	8.8.8.8/24
R3	3.3.3.3/24	R9	9.9.9.9/24
R4	4.4.4.4/24	R10	10.10.10.10/24
R5	5.5.5.5/24	R11	11.11.11.11/24
R6	6.6.6.6/24	R12	12.12.12.12/24

4. Core Concepts and Technologies

4.1 Border Gateway Protocol (BGP)

BGP is a path vector routing protocol standardized in RFC 4271. It is the routing protocol used to exchange routing information between Autonomous Systems on the internet. Unlike interior routing protocols that focus on finding the shortest or fastest path, BGP focuses on policy-based routing, allowing network operators to control which paths traffic takes based on business relationships and routing policies.

BGP operates over TCP port 179 and establishes sessions called BGP neighborships or peering sessions between routers. Once a session is established, routers exchange network reachability information in the form of BGP UPDATE messages containing Network Layer Reachability Information (NLRI) and associated path attributes.

4.2 External BGP (eBGP)

External BGP refers to BGP sessions established between routers in different Autonomous Systems. In this lab, eBGP sessions were configured on all border routers where AS boundaries were crossed. eBGP sessions use the directly connected interface IP address as the neighbor address because the two routers share a physical link across the AS boundary.

A key characteristic of eBGP is that routes learned via eBGP are automatically considered reachable because the neighbor is directly connected. No additional routing configuration is needed to reach eBGP neighbors.

4.3 Internal BGP (iBGP)

Internal BGP refers to BGP sessions established between routers within the same Autonomous System. iBGP is used to propagate external BGP routes learned at the AS border throughout the entire AS so all routers have a complete view of external destinations.

iBGP sessions use loopback addresses as the neighbor address rather than directly connected interface addresses. This provides session stability because a loopback never goes down. For this reason, an additional configuration command is required on iBGP sessions:

```
neighbor X.X.X.X update-source loopback0
```

This tells the router to use its own loopback address as the source of the BGP session rather than the outgoing interface address. Without this command, the iBGP session will not form even if the routers can reach each other.

iBGP also follows a split horizon rule. A router will not advertise routes learned from one iBGP peer to another iBGP peer. This prevents routing loops but means that a full mesh of iBGP sessions is required between all routers in an AS. Every router must peer directly with every other router in the same AS.

4.4 Interior Gateway Protocol (OSPF) as the IGP Foundation

An Interior Gateway Protocol is any routing protocol used to exchange routing information within a single Autonomous System. OSPF (Open Shortest Path First) is the most widely used IGP in enterprise and service provider networks.

In this lab, OSPF served a specific and critical purpose. Because iBGP sessions use loopback addresses, and loopback addresses are virtual interfaces with no physical cable, the routers need a way to reach each other's loopbacks before iBGP sessions can form. OSPF was deployed within each AS to advertise the loopback addresses and all internal link subnets, making every router's loopback reachable by every other router in the same AS.

Without OSPF running first, iBGP sessions would remain in a permanent Idle or Active state because the routers cannot reach the neighbor loopback addresses. The correct operational order is: configure interfaces, configure loopbacks, deploy OSPF, verify loopback reachability, then configure BGP.

Each AS ran its own completely independent OSPF process. The four OSPF processes had no awareness of each other and did not exchange routes across AS boundaries. OSPF routes never crossed the AS boundary. Only BGP carries routes between Autonomous Systems.

4.5 next-hop-self

When a border router learns routes from an eBGP neighbor and redistributes them to its iBGP peers inside the AS, it advertises the original eBGP next-hop IP address along with the route. This is the IP address of the router in the neighboring AS that originated the route.

The problem is that internal routers inside the AS have no route to that external next-hop IP address. OSPF only knows about internal subnets. When an internal router receives the BGP route with an unreachable next-hop, it installs the route in the BGP table but cannot install it in the routing table, meaning traffic will be dropped.

The next-hop-self command solves this problem. Applied to iBGP neighbors on a border router, it instructs the border router to replace the original eBGP next-hop with its own loopback address before advertising the route to its iBGP peers. Since loopback addresses are advertised by OSPF throughout the AS, every internal router can now reach the next-hop and install the route in its routing table.

next-hop-self is applied only on border routers and only toward their iBGP peers. It is not needed for eBGP sessions.

5. Configuration Process

The configuration was executed in five sequential phases. Each phase had to be verified before proceeding to the next to ensure that issues were caught at the correct layer rather than masked by higher-layer problems.

5.1 Phase 1: Physical Interface Configuration

The first phase involved assigning IP addresses to all physical Ethernet interfaces on all 12 routers and bringing them up with the no shutdown command. Physical interfaces in Cisco IOS are administratively shut down by default and must be explicitly activated.

Each router was accessed through the EVE-NG console and configured individually. The interface assignments were cross-referenced with the EVE-NG topology diagram to ensure each IP address was applied to the correct physical interface. A common mistake encountered during this phase was assigning an IP address to the wrong interface, which caused connectivity failures that were not immediately obvious until verified with the show ip interface brief command.

After configuring all interfaces, each direct link was verified with a simple ping to confirm connectivity before proceeding.

5.2 Phase 2: Loopback Interface Configuration

The second phase involved creating loopback interfaces on all 12 routers and assigning them IP addresses using the Y.Y.Y.Y/24 scheme. Unlike physical interfaces, loopback interfaces do not require the no shutdown command as they are active by default upon creation.

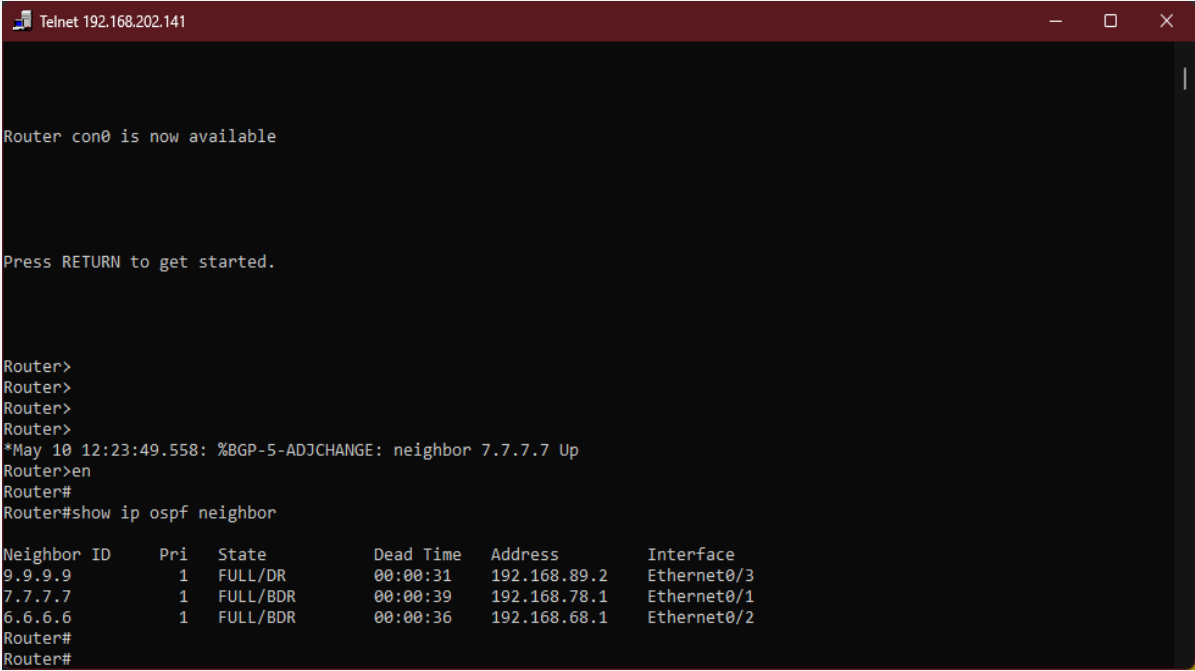
Loopbacks were verified with show ip interface brief to confirm they appeared in the interface table with an up/up status.

5.3 Phase 3: OSPF Configuration

OSPF was configured independently within each of the four Autonomous Systems. Each AS ran its own OSPF process using process ID 1. The router-id was set to the loopback address of each router for consistency and clarity.

The network command was used to advertise subnets into OSPF. For each router, two categories of subnets were advertised: the loopback subnet and all internal link subnets connecting to routers within the same AS. Subnets connecting to routers in a different AS were deliberately excluded from OSPF as those links would be handled by BGP.

OSPF neighbor relationships were verified with `show ip ospf neighbor`. All neighbors were required to reach the FULL state before BGP configuration began. Loopback reachability was then tested with ping commands sourced from the loopback interface to confirm that OSPF was functioning correctly.



```
Telnet 192.168.202.141

Router con0 is now available

Press RETURN to get started.

Router>
Router>
Router>
Router>
*May 10 12:23:49.558: %BGP-5-ADJCHANGE: neighbor 7.7.7.7 Up
Router>en
Router#
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
9.9.9.9        1     FULL/DR         00:00:31    192.168.89.2   Ethernet0/3
7.7.7.7        1     FULL/BDR        00:00:39    192.168.78.1   Ethernet0/1
6.6.6.6        1     FULL/BDR        00:00:36    192.168.68.1   Ethernet0/2
Router#
Router#
```

Figure 2: OSPF Neighbor Adjacencies on R8 (AS64910)

5.4 Phase 4: BGP Configuration

BGP was configured on all 12 routers. Each router was assigned to its Autonomous System with the `router bgp` command followed by the AS number. The `bgp router-id` was set to match the loopback address for consistency.

The configuration followed two distinct rules depending on the type of neighbor relationship.

iBGP Neighbor Configuration

For routers within the same Autonomous System, the `neighbor remote-as` command used the same AS number as the local router, indicating an iBGP relationship. The loopback address of the remote router was used as the neighbor address, and the `update-source loopback0` command was added to instruct the local router to source the BGP session from its own loopback. A full mesh of iBGP sessions was established within each AS, meaning every router peered directly with every other router in the same AS.

eBGP Neighbor Configuration

For routers in different Autonomous Systems, the neighbor remote-as command used the AS number of the neighboring AS, indicating an eBGP relationship. The directly connected interface IP address was used as the neighbor address. No update-source command was needed for eBGP sessions.

Every router also advertised its own loopback subnet into BGP using the network command with the mask keyword. This made each router's loopback reachable from anywhere in the network through BGP.

```
Telnet 192.168.202.141
0;router4
Router>
Router>EN
Router#
Router#SHOW IP BGP SUMMARY
BGP router identifier 4.4.4.4, local AS number 64905
BGP table version is 6, main routing table version 6
7 network entries using 1008 bytes of memory
9 path entries using 756 bytes of memory
5/4 BGP path/bestpath attribute entries using 800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2612 total bytes of memory
BGP activity 7/0 prefixes, 9/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
5.5.5.5        4      64905      7      8        6    0    0 00:00:50      3
192.168.24.1   4      64900      6      8        5    0    0 00:01:34      2
192.168.34.1   4      64900      6      8        5    0    0 00:01:28      3
Router#
Router#
```

Figure 3: BGP Summary on R4 showing both eBGP and iBGP sessions Established

```
Telnet 192.168.202.141
0;router9
Router>
Router>
Router>
Router>EN
Router#
Router#SHOW IP BGP SUMMARY
BGP router identifier 9.9.9.9, local AS number 64910
BGP table version is 8, main routing table version 8
9 network entries using 1296 bytes of memory
14 path entries using 1176 bytes of memory
5/4 BGP path/bestpath attribute entries using 800 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3320 total bytes of memory
BGP activity 9/0 prefixes, 14/0 paths, scan interval 60 secs

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd
6.6.6.6        4    64910      8       8        8    0    0 00:01:07      3
7.7.7.7        4    64910     10      8        8    0    0 00:01:03      3
8.8.8.8        4    64910      6       8        8    0    0 00:01:07      1
192.168.109.2  4    64915      6      10        8    0    0 00:01:12      3
192.168.119.2  4    64915      6      10        8    0    0 00:02:00      3
Router#
Router#
Router#
Router#
Router#
```

Figure 4: BGP Summary on R9 showing all five neighbor sessions Established

5.5 Phase 5: next-hop-self

After BGP sessions were established, end-to-end ping tests failed despite all BGP neighbors showing as Established. The cause was identified as the next-hop problem. Border routers were advertising external routes to their iBGP peers with the original eBGP next-hop address, which was an external IP address unreachable by internal routers.

The next-hop-self command was applied on all border routers toward their iBGP peers only. This covered R2, R3, R4, R5, R6, R7, R9, R10, and R11. Purely internal routers (R1, R8, R12) did not require this command as they had no eBGP connections and therefore never introduced external next-hops into the iBGP.

After applying next-hop-self, the routing tables were updated and end-to-end pings succeeded.

6. Verification and Results

6.1 OSPF Verification

OSPF neighbor adjacencies were verified on all routers using `show ip ospf neighbor`. All expected neighbor relationships reached the FULL state, confirming that the IGP was functioning correctly within each AS. The key verification was that loopbacks were reachable across each AS using ping with the source `loopback0` option.

6.2 BGP Session Verification

BGP neighbor sessions were verified on all 12 routers using `show ip bgp summary`. The `State/PfxRcd` column was used to determine session status. A numeric value in this column indicates the session is Established and the number represents the count of prefixes received from that neighbor. Sessions showing Idle or Active indicated a problem requiring investigation.

All 12 routers achieved full BGP session establishment. The total number of iBGP and eBGP sessions across the lab was as follows:

- AS64900 had iBGP sessions between R1, R2, and R3 (partial mesh based on physical connectivity) plus eBGP from R2 and R3 to R4.
- AS64905 had one iBGP session between R4 and R5, plus eBGP from R4 to R2 and R3, and eBGP from R5 to R6 and R7.
- AS64910 had a full iBGP mesh between R6, R7, R8, and R9, plus eBGP from R6 and R7 to R5, and eBGP from R9 to R10 and R11.
- AS64915 had iBGP sessions between R10, R11, and R12, plus eBGP from R10 and R11 to R9.

```
Telnet 192.168.202.141
*May 9 15:26:37.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
*May 9 15:26:37.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/2, changed state to down
*May 9 15:26:37.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3, changed state to down
*May 9 15:26:38.084: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*May 9 15:26:39.112: %LINK-5-CHANGED: Interface Ethernet0/2, changed state to administratively down
*May 9 15:26:39.112: %LINK-5-CHANGED: Interface Ethernet0/3, changed state to administratively down
*May 9 15:26:42.712: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Startup Config Present)
Router>
Router>
Router>
Router>EN
Router#
Router#
Router#
Router#SHOW IP BGP SUMMARY
BGP router identifier 1.1.1.1, local AS number 64900
BGP table version is 1, main routing table version 1
1 network entries using 144 bytes of memory
1 path entries using 84 bytes of memory
1/0 BGP path/bestpath attribute entries using 160 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 388 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
2.2.2.2       4      64900      0       0        1    0    0 never    Idle
3.3.3.3       4      64900      0       0        1    0    0 never    Idle
Router#
Router#
```

Figure 5: BGP Summary on R1 confirming all sessions Established

6.3 Routing Table Verification

The BGP routing table was examined on R1 using show ip route bgp. This confirmed that R1 had learned routes to all 12 loopback subnets across all four Autonomous Systems through BGP. The presence of these routes in the routing table confirmed that next-hop-self was functioning correctly and that the routes were installable.

```
Telnet 192.168.202.141
0;router1
Router#
Router#SHOW IP ROUTE BGP
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

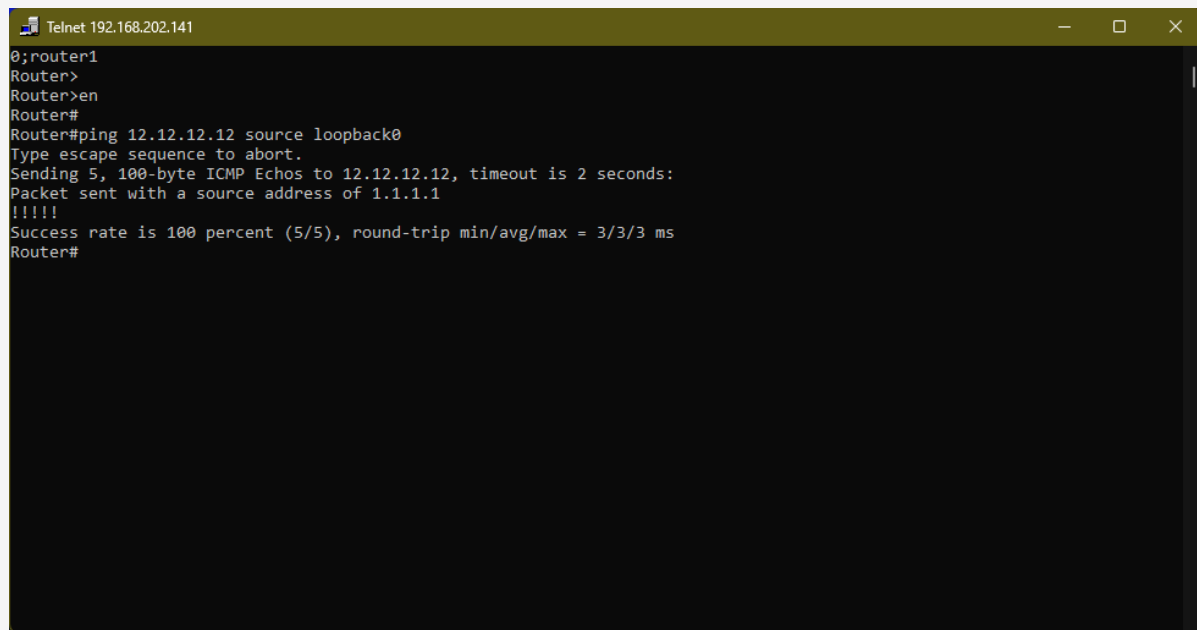
Gateway of last resort is not set

    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
B       2.2.2.0/24 [200/0] via 2.2.2.2, 00:04:00
    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
B       3.3.3.0/24 [200/0] via 3.3.3.3, 00:04:00
    4.0.0.0/24 is subnetted, 1 subnets
B       4.4.4.0 [200/0] via 192.168.24.2, 00:03:58
    5.0.0.0/24 is subnetted, 1 subnets
B       5.5.5.0 [200/0] via 192.168.24.2, 00:03:27
Router#
Router#
```

Figure 6: BGP Routes in R1 Routing Table

6.4 End-to-End Connectivity Verification

The final verification was an end-to-end ping from R1 in AS64900 to R12 in AS64915 using the loopback interface as the source. This ping traversed the full path across all four Autonomous Systems and confirmed that the complete BGP network was operational.

A screenshot of a Telnet window titled 'Telnet 192.168.202.141'. The window has a dark background with white text. The text shows a sequence of commands and responses in a network configuration environment. The user enters '0;router1', then 'Router>', then 'Router>en', then 'Router#'. The user then enters 'Router#ping 12.12.12.12 source loopback0'. The system responds with 'Type escape sequence to abort.', 'Sending 5, 100-byte ICMP Echos to 12.12.12.12, timeout is 2 seconds:', 'Packet sent with a source address of 1.1.1.1', '!!!!!!', and 'Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/3 ms'. The prompt returns to 'Router#'.

```
0;router1
Router>
Router>en
Router#
Router#ping 12.12.12.12 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.12.12.12, timeout is 2 seconds:
Packet sent with a source address of 1.1.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/3 ms
Router#
```

Figure 7: End-to-End Ping from R1 (AS64900) to R12 (AS64915)

A 100 percent success rate on this ping confirmed that all configuration phases were completed correctly and the entire multi-AS BGP network was functioning as intended.

7. Challenges Encountered and Lessons Learned

7.1 Wrong Interface IP Assignments

During Phase 1, IP addresses were configured on incorrect physical interfaces on one of the routers. This caused OSPF and BGP sessions on that router to fail to form because the IPs were applied to interfaces connected to the wrong neighbors. The error was identified by running `show ip interface brief` and comparing the assigned IPs against the topology diagram, then pinging across each direct link to isolate which link was broken. The lesson learned was that every direct link should be verified with a ping immediately after interface configuration, before moving to any higher-layer protocol.

7.2 IOS show Command Case Sensitivity

The IOS pipe filter using the `section` keyword is case sensitive. Commands typed in uppercase such as `SHOW RUNNING-CONFIG PIPE SECTION BGP` returned empty output because the stored configuration uses lowercase keywords. All show commands with pipe filters must use lowercase section names. This was a small but time-consuming issue that repeated itself several times during the lab.

7.3 OSPF Configuration Not Saved

On several routers, OSPF configuration appeared to be entered correctly, but subsequent `show ip ospf neighbor` output returned empty results. The cause was that the configuration had not been saved with the `wr` command, and the running configuration had been lost. The habit of saving with `wr` immediately after completing configuration on each router was established as a standard practice for the remainder of the lab.

7.4 BGP Sessions Established But Traffic Still Failing

After all BGP sessions showed as Established, end-to-end ping tests failed completely. This was an instructive moment because the assumption was that Established BGP sessions meant the network was working. The investigation revealed the next-hop problem. BGP sessions being Established only means the routers have successfully exchanged routing information. It does not mean the routes are actually usable. The routes must have a reachable next-hop IP for them to be installed in the routing table and used to forward traffic. This distinction between BGP session state and route usability is a fundamental concept in BGP troubleshooting.

7.5 update-source loopback0 Requirement for iBGP

Several iBGP sessions remained in the Active state even after OSPF was confirmed to be advertising the loopback addresses correctly. The cause was a missing `update-source loopback0` command. Without this command, the router attempts to source the BGP session from the outgoing physical interface rather than the loopback. The remote router receives a TCP connection from an unexpected source IP and rejects it, keeping the session in the Active state. The `update-source` command must accompany every iBGP neighbor statement.

8. Conclusion

This lab successfully demonstrated the configuration and operation of a multi-Autonomous System BGP network in EVE-NG. Starting from a blank topology, all 12 routers were configured from the physical interface layer through to full BGP route exchange and verified with successful end-to-end connectivity across all four Autonomous Systems.

The lab reinforced several important principles that apply directly to real-world network engineering. The layered approach to configuration, starting with physical connectivity, then IGP, then BGP, mirrors the troubleshooting methodology used in production networks. The distinction between iBGP and eBGP, the role of OSPF as the IGP foundation for iBGP, and the necessity of next-hop-self are all concepts that appear in Cisco CCNA, CCNP, and real-world BGP deployments.

This lab forms part of an ongoing network security engineering portfolio documenting hands-on experience with enterprise networking, routing protocols, firewall administration, and security operations, building toward CCNP Security certification and a career in network security engineering.

Timothy Otim Lutara

github.com/Otim24/nsec-portfolio | linkedin.com/in/timothy-otim-lutara